

AWS MLOps Project to Deploy a Classification Model [Banking]

Components:

- AWS S3
- AWS EKS
- AWS ECR
- AWS CodeBuild
- AWS CodePipeline
- Application Load Balancer
- AWS CloudWatch

Note:

The Cost Breakdown as of November 4, 2025. The costs might get updated by AWS Cloud as per their latest pricing change. The pricing may vary depending on the AWS region where the services are created. We recommend visiting the official AWS pricing page to check the current cost of the services to come up with updated guesstimates.

AWS Services Cost: <https://aws.amazon.com/pricing>

Cost Breakdown:

Components	Description	Usage	Estimated Cost
Storage	AWS S3 to Store Data Files and Codepipeline Artifacts	Example: 10 GB total (data files + artifacts)	~\$0.23 / month (\$0.023 × 10 GB)
Docker Image Storage	AWS ECR to store Docker Images	3–5 images (~500 MB each)	~\$0.05 / month (minimal storage + free first 500 MB)
Containerization	AWS Codebuild to build and package Docker images	~100 build minutes per month using arm1.medium	~\$0.50 / month (100 × \$0.005)
Deployment	AWS CodePipeline automates the CI/CD deployment process	One active pipeline connected to a GitHub repository	~\$1.00 / month
Compute	AWS EKS to	• 1 Standard	~\$132/month

Resources	orchestrate containerized applications	EKS Cluster running - Cluster control plane fee: \$0.10 per hour. Worker Nodes (EC2 instances) - e.g., 2 × t3.medium nodes (2 vCPU, 4 GB RAM each)	
Load Balancing	Application Load Balancer (ALB) routes incoming API requests to ECS containers	24×7 uptime with light API traffic (~1 GB processed)	~\$18.00 / month
Monitoring	AWS CloudWatch collects logs and metrics	5 GB of logs + 100 custom metrics	~\$5.00 / month

Total Estimated Cost: If a Flask-based ML API container on AWS EKS with CI/CD is running after the deployment via CodePipeline and CodeBuild:

Cost = (S3 \$0.008 + ECR \$0.0016 + CodeBuild \$0.016 + CodePipeline \$0.033 + EKS \$4.40+ ALB \$0.60 + CloudWatch \$0.167 + Data Transfer \$0.03)
= **~\$5.26/day (approximate cost)**

Note:

- We recommend monitoring AWS Services and deleting those that are not in use after project execution, whether they were created using the UI or Terraform.
- These are purely estimated costs of the services. The actual charges may vary depending on several factors, such as usage duration, service configuration, and region. Please note that these are only estimates; the actual cost will depend on how the services are utilized.